EUCLID · BOOK I · PROPOSITION 16

Put an interior angle inside an exterior angle

Bisect a side, extend a joining line, and use vertically opposite angles to copy an interior angle into the exterior. The whole is larger than its proper part. Repeat for the other interior angle.

THE PROPOSITION

GREEK Ἑλληνικά

Παντὸς τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν.

LATIN Latine

In quouis triangulo uno latere producto angulus extrinsecus positus utrouis angulo interiore et opposito maior est.

ENGLISH reading translation

If one side of any triangle is extended, the exterior angle is greater than either of the interior and opposite angles.

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1 Extend ΒΓ past Γ to Δ

GREEK Ἑλληνικά

Ἔστω τρίγωνον τὸ ΑΒΓ, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἡ ΒΓ ἐπὶ τὸ Δ· λέγω, ὅτι ἡ ἐκτὸς γωνία ἡ ὑπὸ ΑΓΔ μείζων ἐστὶν ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον τῶν ὑπὸ ΓΒΑ, ΒΑΓ γωνιῶν.

LATIN Latine

Sit triangulus ΑΒΓ, et producatur unum latus eius ΒΓ ad Δ punctum. dico esse ∠ ΑΓΔ > ΓΒΑ et ΑΓΔ > ΒΑΓ.

ENGLISH reading translation

Let ΑΒΓ be a triangle and extend its side ΒΓ to Δ. I claim that exterior angle ΑΓΔ is greater than each opposite interior angle ΓΒΑ and ΒΑΓ.

2 Bisect AΓ at E

GREEK Ἑλληνικά

Τετμήσθω ἡ ΑΓ δίχα κατὰ τὸ Ε,

LATIN Latine

secetur AΓ in duas partes aequales in E,

ENGLISH reading translation

Bisect AΓ at E.

3 Join, extend, copy, and join again

GREEK Ἑλληνικά

καὶ ἐπιζευχθεῖσα ἡ ΒΕ ἐκβεβλήσθω ἐπ᾽ εὐθείας ἐπὶ τὸ Ζ, καὶ κείσθω τῇ ΒΕ ἴση ἡ ΕΖ, καὶ ἐπεζεύχθω ἡ ΖΓ, καὶ διήχθω ἡ ΑΓ ἐπὶ τὸ Η.

LATIN Latine

et ducta BE producatur in directum ad Z, et ponatur EZ = BE, et ducatur ZΓ, et educatur AΓ ad H.

ENGLISH reading translation

Join BE and extend it straight to Z, laying off EZ equal to BE. Join ZΓ and continue AΓ to H.

4 Collect the two equal side pairs

GREEK Ἑλληνικά

Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν ΑΕ τῇ ΕΓ, ἡ δὲ ΒΕ τῇ ΕΖ, δύο δὴ αἱ ΑΕ, ΕΒ δυσὶ ταῖς ΓΕ, ΕΖ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ·

LATIN Latine

iam quoniam AE = EΓ et BE = EZ, duae rectae AE, EB duabus ΓE, EZ aequales sunt altera alteri.

ENGLISH reading translation

AE equals EΓ, and BE equals EZ. Thus the two sides AE, EB equal ΓE, EZ respectively.

5 Use I.15 at the crossing E

GREEK Ἑλληνικά

καὶ γωνία ἡ ὑπὸ ΑΕΒ γωνίᾳ τῇ ὑπὸ ΖΕΓ ἴση ἐστίν· κατὰ κορυφὴν γάρ·

LATIN Latine

et ∠ AEB = ZEΓ (nam ad uerticem eius est).

ENGLISH reading translation

Angle AEB equals ZEΓ, for they are vertically opposite.

6 Apply side–angle–side

GREEK Ἑλληνικά

βάσις ἄρα ἡ ΑΒ βάσει τῇ ΖΓ ἴση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΕΓ τριγώνῳ ἐστὶν ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, ὑφ᾽ ἃς αἱ ἴσαι πλευραὶ ὑποτείνουσιν·

LATIN Latine

itaque basis AB basi ZΓ aequalis est et △ ABE = ZEΓ, et reliqui anguli reliquis aequales sunt alter alteri, sub quibus aequalia latera subtendunt.

ENGLISH reading translation

Therefore AB equals ZΓ, the triangles ABE and ZEΓ are equal, and their remaining angles are equal respectively where the equal sides subtend them.

7 Identify the copied interior angle

GREEK Ἑλληνικά

ἴση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΕ τῇ ὑπὸ ΕΓΖ.

LATIN Latine

itaque ∠ BAE = EΓZ.

ENGLISH reading translation

In particular, BAE equals EΓZ.

8 The exterior angle contains that angle as a proper part

GREEK Ἑλληνικά

μείζων δέ ἐστιν ἡ ὑπὸ ΕΓΔ τῆς ὑπὸ ΕΓΖ· μείζων ἄρα ἡ ὑπὸ ΑΓΔ τῆς ὑπὸ ΒΑΕ.

LATIN Latine

uerum ∠ ΕΓΔ > ΕΓΖ. quare ∠ ΑΓΔ > ΒΑΕ.

ENGLISH reading translation

But ΕΓΔ is greater than its part ΕΓΖ. Therefore exterior angle ΑΓΔ is greater than ΒΑΕ.

9 Bisect ΒΓ and repeat for the other interior angle

GREEK Ἑλληνικά

Ὁμοίως δὴ τῆς ΒΓ τετμημένης δίχα δειχθήσεται καὶ ἡ ὑπὸ ΒΓΗ, τουτέστιν ἡ ὑπὸ ΑΓΔ, μείζων καὶ τῆς ὑπὸ ΑΒΓ.

LATIN Latine

similiter recta ΒΓ in duas partes aequales secta demonstrabitur etiam ∠ ΒΓΗ > ΑΒΓ, h. e. ∠ ΑΓΔ > ΑΒΓ.

ENGLISH reading translation

Similarly, after bisecting ΒΓ, ΒΓΗ will be shown greater than ΑΒΓ too. That is, ΑΓΔ is greater than ΑΒΓ.

10 The exterior is greater than each opposite interior angle

GREEK Ἑλληνικά

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

LATIN Latine

Ergo in quouis triangulo uno latere producto angulus extrinsecus positus utrouis angulo interiore et opposito maior est; quod erat demonstrandum.

ENGLISH reading translation

Therefore, when one side of any triangle is extended, the exterior angle is greater than each of the opposite interior angles. This is what had to be shown.

Stoicheia · Euclid, Elements I.1–16. Greek and Latin: J. L. Heiberg, Euclidis opera omnia, vol. I (Leipzig, 1883), pp. 2–45; public domain. Latin is Heiberg’s nineteenth-century translation. English reading and notes: Stoicheia, 2026, MIT. Source and edition notes: antonsoo.github.io/stoicheia/